

Evaluation of RGB, Depth, and RGB-D Methods for Low-Lying Obstacle Detection in Indoor Robot Navigation

Bibliography Report

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I. INTRODUCTION

Safe navigation in indoor environments requires mobile robots to detect not only large obstacles such as walls and furniture, but also very small and low-lying objects such as cables, latex gloves, screws, and other floor-level hazards. These obstacles are particularly challenging because they occupy few pixels in images captured by the robot's camera and may also be difficult to distinguish from the floor using depth sensors. Previous work has therefore explored both appearance-based methods using RGB images and geometric methods using depth or RGB-D information. Hua et al. showed that combining RGB and depth information can improve small-obstacle segmentation for robotic navigation, although the obstacles considered in their experiments were generally larger than very thin or flat objects such as cables or screws [1].

The current objective of this project is to evaluate methods for detecting thin and low-lying indoor obstacles, with particular emphasis on the contribution of RGB and stereo-depth information. The methods will first be evaluated using the Indoor Small Object Dataset (ISOD), which provides aligned RGB and depth data together with binary semantic segmentation labels. Candidate semantic segmentation models include established architectures such as SegNet and DeepLabV3+, as well as lightweight models suitable for real-time robotic applications. The most promising approaches may subsequently be tested using images acquired with the robot's camera in real indoor environments.

II. STATE OF THE ART

Hua et al. [1] proposed a two-stage RGB-D semantic segmentation system for small-obstacle avoidance. The first stage identifies the traversable road or floor region, while the second stage detects obstacles inside this region. Separate RGB and depth branches are used to extract and combine visual and geometric information. Their work demonstrates the usefulness of RGB-D sensing for obstacle perception, but their custom dataset mainly contains obstacles substantially larger than very thin floor hazards such as cables.

Chen et al. [2] proposed MOSTS, a lightweight one-shot texture segmentation model for indoor robot navigation. Their

system first estimates the ground plane from depth, point-cloud, and IMU information using filtering, RANSAC, and PCA, and then applies RGB texture segmentation to identify traversable floor regions. The authors report that their geometric ground-plane method typically detects objects higher than approximately 5 cm, meaning that very low objects may be missed by depth sensing. MOSTS addresses this limitation by comparing a query image with a reference image of the desired floor texture. On ISOD, MOSTS with a MobileNetV3 backbone achieves 85.7% mIoU while remaining suitable for real-time operation. However, its dependence on a reference floor texture may reduce robustness when several valid floor appearances occur in the same environment.

Semantic segmentation architectures provide another direct approach to the problem because they classify each image pixel rather than requiring individual obstacle instances. SegNet uses an encoder-decoder architecture in which pooling indices from the encoder are reused during upsampling, providing an efficient approach to pixel-wise semantic segmentation [3]. DeepLabV3+ combines atrous spatial pyramid pooling with an encoder-decoder architecture to capture multi-scale contextual information while improving object boundaries [4]. These architectures are relevant candidate baselines because the ISOD ground truth is formulated as a binary semantic segmentation problem.

For deployment on mobile robots, computational efficiency is also important. Fast-SCNN was specifically designed for fast semantic segmentation on embedded devices with limited memory. It combines spatial information at relatively high resolution with semantic features extracted at lower resolution [5]. Such lightweight architectures, as well as efficient backbones such as MobileNet, are potentially attractive for this project because segmentation performance must be balanced with inference speed and computational requirements on robotic hardware.

Caporali et al. [6] presented RT-DLO, a real-time instance-segmentation method for deformable linear objects such as cables, wires, and ropes. The method exploits the linear structure of these objects to separate individual instances, including at intersections. Although it highlights the challenges involved in segmenting thin cable-like objects, RT-DLO is primarily designed for robotic manipulation and instance separation

rather than traversability estimation, and it operates on RGB images without depth information.

Kirillov et al. [7] introduced the Segment Anything Model (SAM), a promptable foundation model capable of producing segmentation masks from prompts such as points, bounding boxes, and masks. SAM may be useful as an annotation tool or as a zero-shot segmentation baseline. However, fine structures and small disconnected components remain challenging cases, which is particularly relevant for thin floor obstacles. Its computational requirements also make lightweight semantic segmentation approaches attractive for real-time robotic deployment.

Together, these studies indicate that detecting very small and low-lying obstacles remains challenging. Depth-based approaches provide useful geometric information but may fail when objects lie very close to the floor, while RGB-based approaches can exploit appearance differences but may struggle with low contrast, fine structures, or changing floor textures. At the same time, real-time robotic deployment requires limited computational cost. Therefore, this project will evaluate semantic segmentation approaches using RGB, depth, and RGB-D information on ISOD and investigate lightweight models suitable for deployment on a mobile robot. The most promising methods will also be considered for testing with the robot's camera in real indoor conditions. This comparison will help determine whether stereo depth provides a measurable advantage for detecting thin and low-profile obstacles and which approaches provide an appropriate balance between segmentation performance and computational efficiency.

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